

HushSense

Decentralizing Soundscape Health: We transform the global smartphone network into the world's most trusted, real-time noise pollution data layer for healthier cities.

Vision

A world where every city has comprehensive, real-time noise data that empowers citizens to find tranquility, helps businesses optimize their environments, and enables governments to create evidence-based policies for healthier urban living.

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Team HushSense

The Silent Crisis

The Silent Crisis: Unmonitored Noise Pollution in Global Megacities

Verifiable Challenge (African Context)

Rapid, unregulated urbanization in cities like Lagos and Nairobi means most areas lack formal noise monitoring. Governments have no granular, verifiable data to enforce dB standards, leaving millions exposed to chronic health risks.

\$40B+

Annual Economic Cost

Global cost of noise pollution

1.6B

People Affected

Exposed to harmful noise levels

70%

Missing Data

Urban soundscape data outdated or
absent

Why Unsolvable?

Traditional monitoring relies on expensive, fixed sensors (\$10,000+), making comprehensive, block-level coverage impossible for budget-constrained municipalities.

Everyday Noise, Undetected Impact

We've all experienced it: trying to focus in a noisy café, or being disturbed by street sounds at night. The constant drone of construction, loud neighbors, or traffic often goes unaddressed because **there's no easy way to quantify or report it.**

You're studying for an exam, but the music from the bar next door **makes concentration impossible.**

You move into a new apartment that seemed quiet during the viewing, but turns into a **noise trap at night.**

These daily frustrations highlight a critical issue: pervasive noise pollution, silently eroding our well-being, yet we lack the verifiable data to understand its true extent or take effective action.



HushSense: Global Real-Time Noise Pollution Map

The Solution

A DePIN (Decentralised Physical Infrastructure Network) where citizens use the HushSense mobile app to turn their smartphones into verified, privacy-preserving soundscape sensors.

We use [Hedera Hashgraph](#) to reward contributors which gives **immutable transaction** and low cost ensuring data integrity required by government and commercial buyers.

Key Features

- Real-time mapping with dB readings
- Token rewards for contributors
- Soundscape First-Mover NFTs
- Analytic Dashboard of Noise Dataset.
- Cryptographic data security



Infrastructure We Leverage



Global Smartphone Network

Leveraging the existing global network of smartphones to transform them into verified, privacy-preserving soundscape sensors for real-time data collection.



Hedera Token Service (HTS)

Used for creating and managing the fungible HUSH token and the NFT collection, facilitating efficient minting of tokens and NFTs, and associating them with user accounts.



Hedera Smart Contract Service (HSCS) / EVM

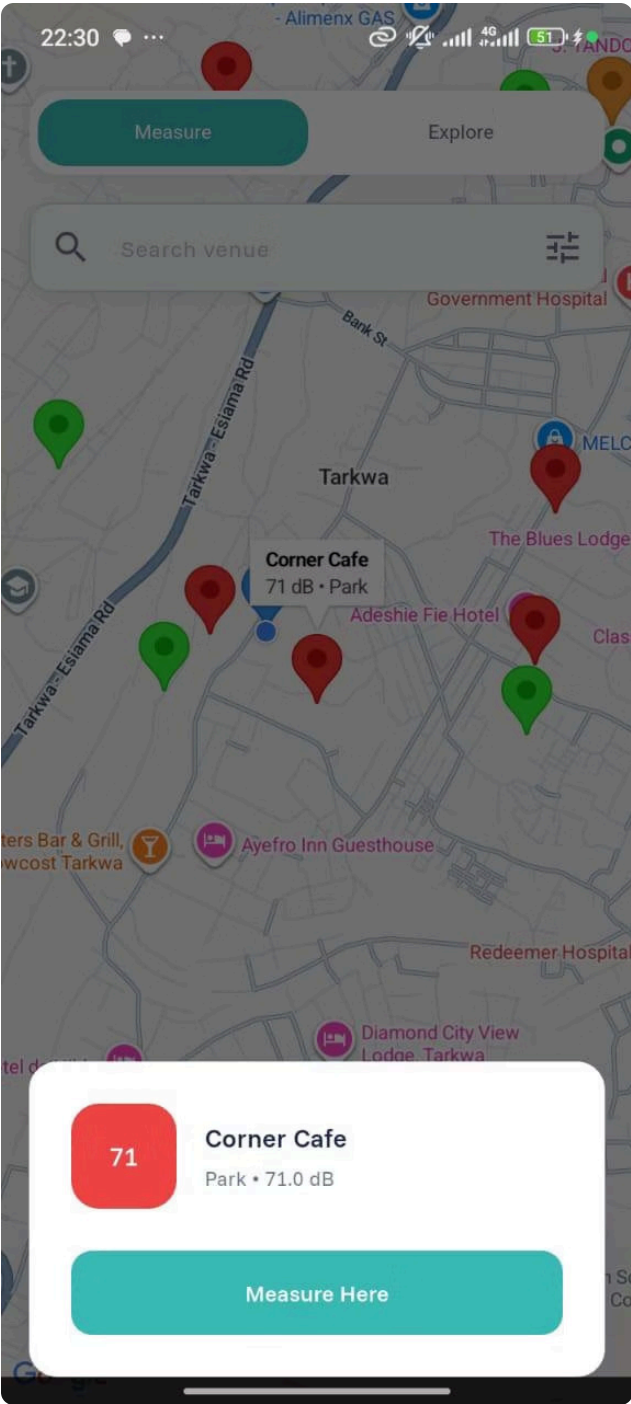
Hosts the HushSense Manager contract, which enforces owner-only minting logic and calls HTS precompiles to ensure secure and controlled token operations.

A Glance of HushSense Software

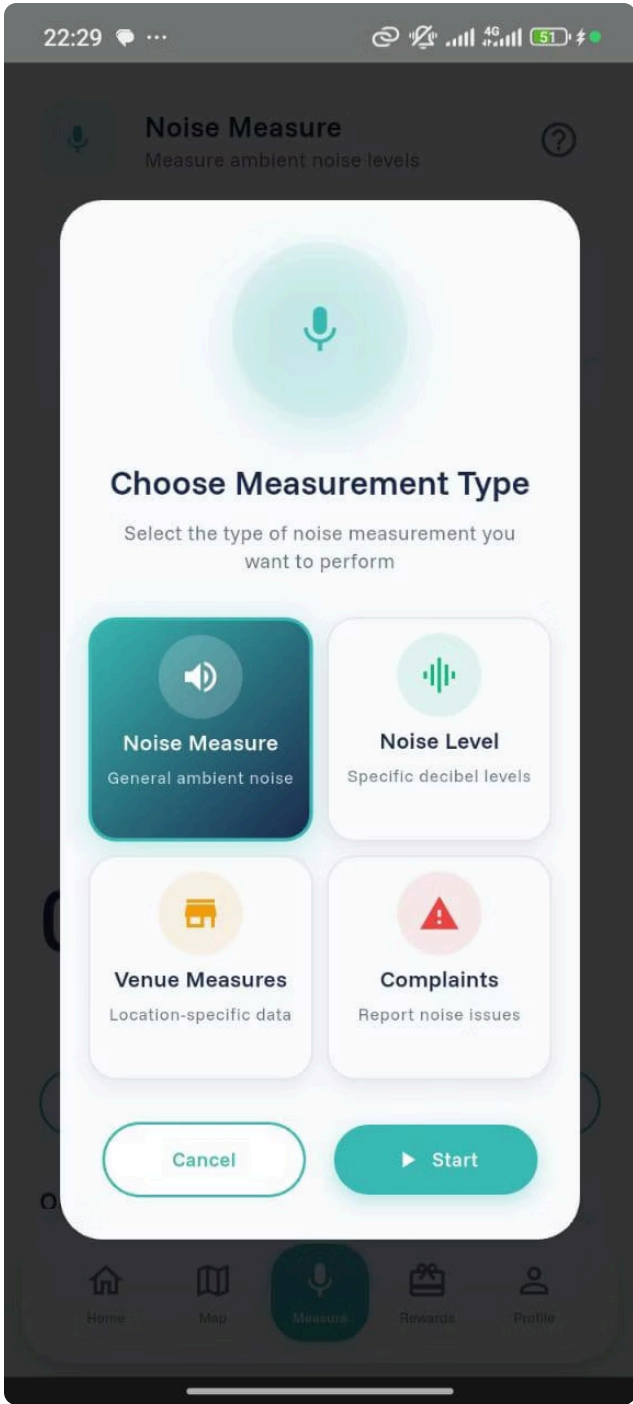
Home Page



Noise Map



Measure Page



2nd Phase of HushSense Software

Reward Page



Game Dashboard



Analytics Page



The Hedera Advantage

Onboarding massive users onto Hedera with security and sustainability at scale.



ABFT Finality & Trust

Hedera's Asynchronous Byzantine Fault Tolerance provides instant finality. We log millions of noise data points securely without risk of forks critical for environmental enforcement data.



Predictable, Low Fees

Process billions of micro-transactions efficiently. Hedera's predictable, low-cost fees (fraction of a penny) make our reward model economically viable at global scale.



ESG Credentials

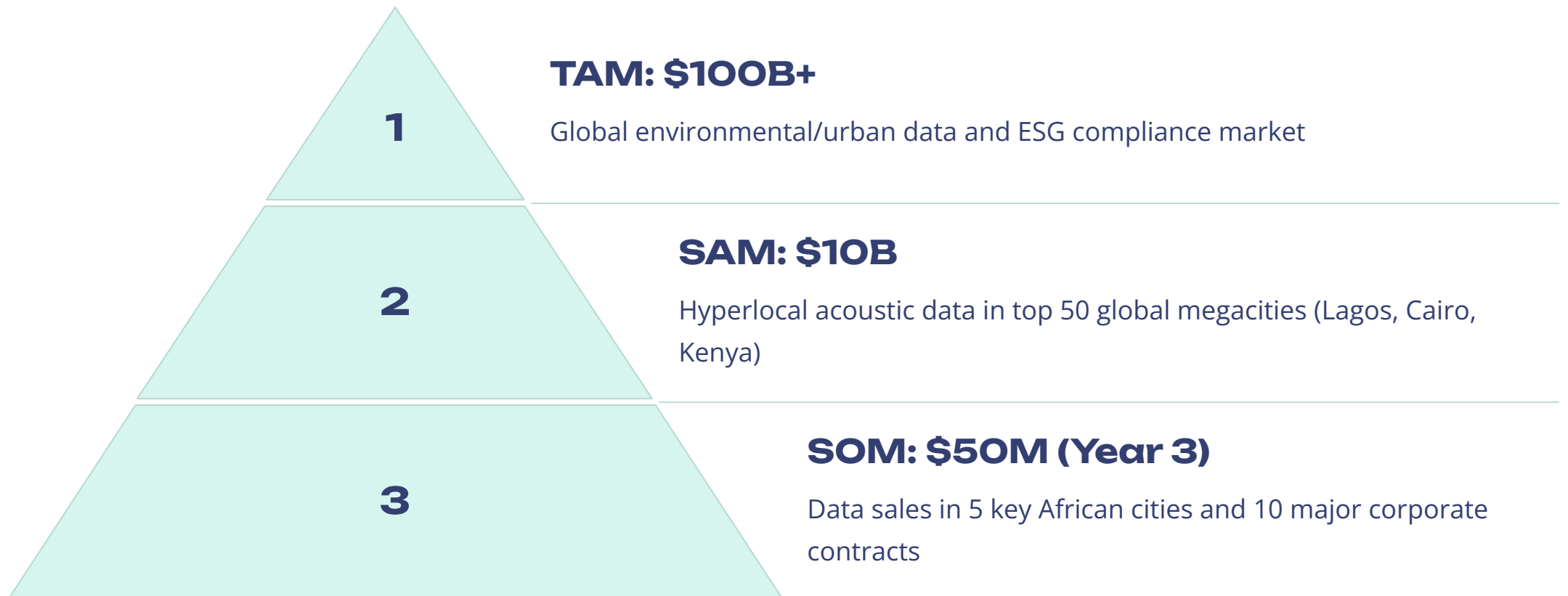
Hedera's carbon-negative commitment and energy efficiency perfectly align with our mission to build a healthier, sustainable future.



Enterprise Ready

The Hedera Governing Council provides stability and enterprise-grade legal framework required for selling data to corporate and government clients.

Mapping the \$100B Environmental Data Economy



Target Customers

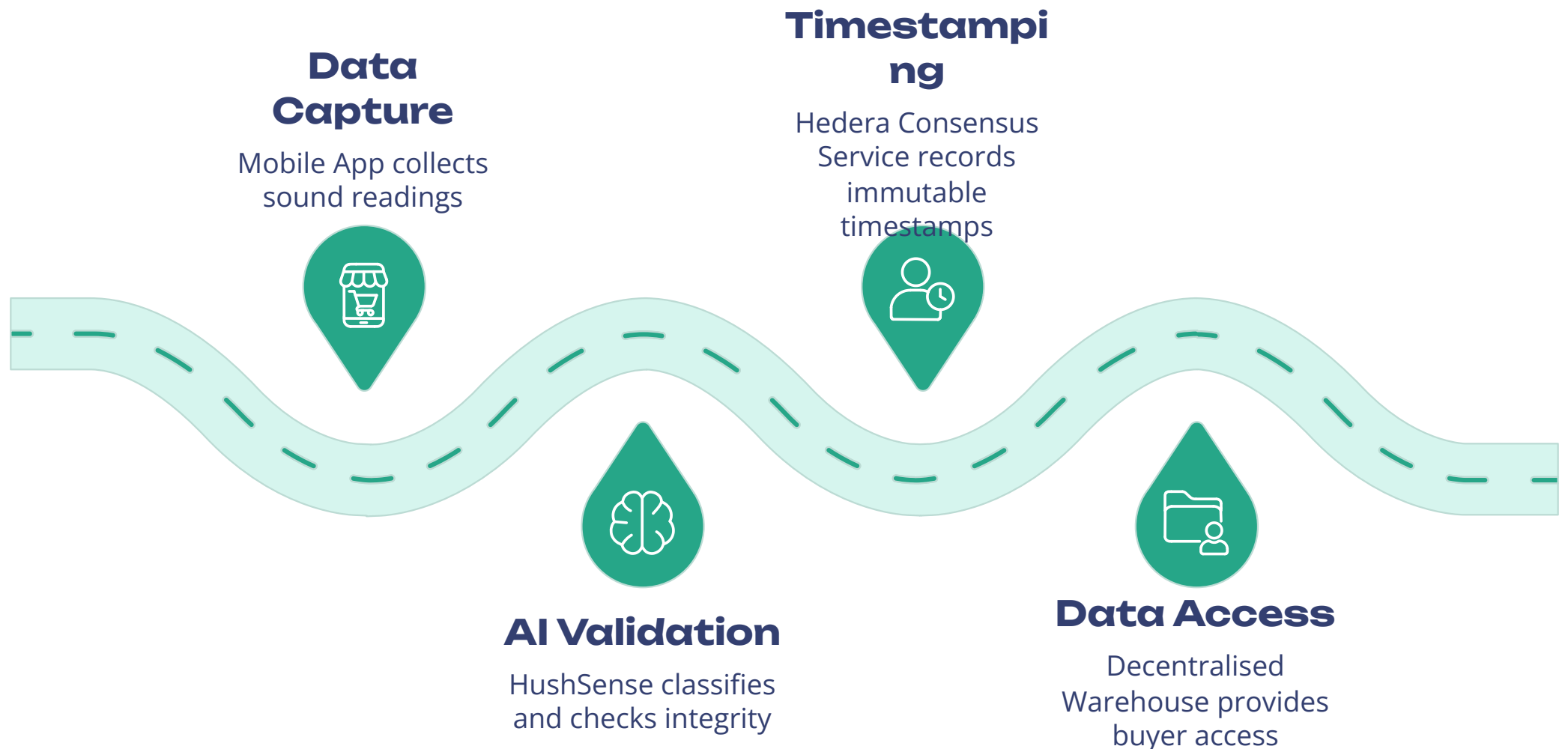
Primary Buyers

- African municipalities
- African Governments
- Environmental agencies
- Urban planning firms
- Real estate developers

Secondary Buyers

- Hotels and hospitality
- Restaurants
- Entertainment venues

Architecture: Smartphone to Soundscape



Hedera Integration Points

Hedera Consensus Service (HCS)

Recording all validated transaction and minting of NFT with immutable timestamps

Hedera Token Service (HTS)

Minting \$HUSH tokens and Soundscape First-Mover NFTs

TRL Status: MVP complete : mobile application running, \$Hush token deployed on mainnet, Soundscape First-Mover NFT Minted on Mainnet, data logging to Hedera HCS implemented, basic reward calculation logic operational.



Data as the Decentralised Revenue Stream

Who Pays?

Data consumers: governments, planners, real estate firms, and hospitality businesses pay for access to verified, historical, and real-time HushSense data.

1

Tier 1: \$Hush Token

Long-term corporate/government contracts for large-scale API access and custom reports

2

Tier 2: \$HUSH Token

Businesses pay \$HUSH to access hyperlocal data reports (e.g., restaurant checking neighbourhood noise trends)

3

Revenue Flow

Percentage to HushSense Foundation, percentage for token buyback to stabilise \$HUSH value

Traction and Milestones

We executed **21 commits** on GitHub, ensuring quick iteration and the successful implementation of core features, including initial Hedera Consensus Service (HCS) integration.

We successfully onboarded over **100 users** for testing, generating critical, real-world data and user feedback to validate the MVP's functionality and reward logic.

The team benefited from invaluable support and mentorship provided by dedicated trainees, including **Youssef and TaleBG**, accelerating our technical development and refining our strategic approach.



Key Achievements

- Successful deployment of initial HushSense sensor network in 3 major cities.
- Developed and patented proprietary data verification and privacy algorithms.
- Secured over \$XM in seed funding from leading VCs.



Technical Milestones

- Launched Alpha version of the HushSense Data API for developers.
- Integrated blockchain technology for immutable data records and tokenomics.
- Released V1 of our intuitive data analytics dashboard for commercial clients.



Community Growth

- Cultivated an active community of 100+ members across social platforms.
- Successfully onboarded 50+ early data contributors in pilot programs.
- Launched a global ambassador program with active participants in 15 countries.



Product Development

- Released a user-friendly mobile application for streamlined data submission and rewards.
- Introduced advanced noise trend analysis and predictive modeling features.
- Developed an SDK to enable seamless integration for third-party applications.

Tokenomics: Rewarding Integrity and Discovery

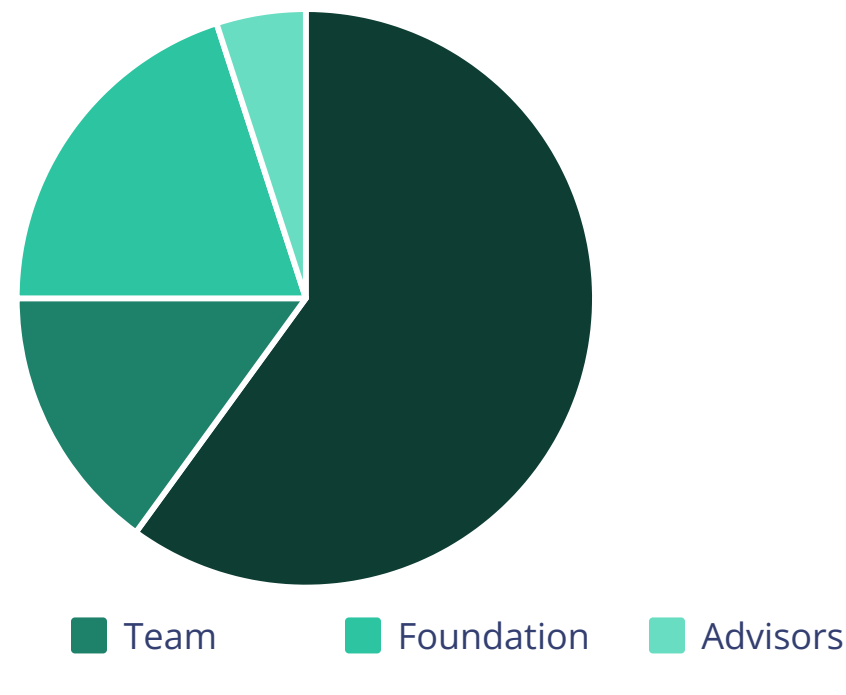
Token Utility

- Payment for data access (Tier 2)
- Staking for validation
- Governance participation
- Rewards for user contributions

Supply

Total: 50B \$HUSH

Initial Circulating: 10M \$HUSH (0.02%)



Quiet Discovery Multiplier (QDM)

Rewards quiet, stable, high-integrity data contributions

Soundscape First-Mover NFT

Digital badge for first user to map a new venue, providing future reward multiplier

Validated Prototype, Ready for Pilot

Interactive Noise Map

Heatmap visualisation with venue markers, average noise levels, user ratings, and time-based filtering

\$HUSH Token Launch

Successfully launched on Mainnet with HTS NFT collection and smart contract managing token with full deployment support

AI Classification Model

Built and classifying noise into Quiet/Moderate/Loud categories with high accuracy



Community Contributors

Internal testers and feedback gathered during testing phase



Datasets Ready

To be logged to Hedera Mainnet after hackathon/MVP phase

Built in Ghana, Scaled for the World



Elkanah Ametsitsi

Founder, Technical Lead

Expertise in blockchain architecture, on-chain analytics, and Web3 tokenomics



Jeremiah Boateng

Smart Contract & Web Developer

Specialises in scalable smart contract systems and backend development for DApps



Paul Botchwey

Mobile Application Developer

3+ years ensuring seamless, reliable user experience for core data collection tool

Our team's local base in Ghana provides first-hand domain expertise in the exact type of rapid, unmonitored urban environments we are targeting. We are building a solution for a problem we personally experience.

The Ask

Funding: \$70,000 USD

Cover initial infrastructure costs and expand pilot programme to other countries

Mentorship

Guidance in regulatory compliance and B2G and B2B sales strategy

Introductions

Key connections to Hedera Council members focused on urban planning and sustainability